AFN-1233

PCA131

P.G.D.C.A. (P.G. DIPLOMA) EXAMINATION DECEMBER 2010

First Semester

Computer Applications

FUNDAMENTALS OF DIGITAL COMPUTER

(Non-CBCS)

Duration : 3 Hours Maximum : 100 Marks

Part - A (15 × 1 = 15)

Answer **all** questions.

1. Define Flow Chart.

2. What is nibble ?

3. Define ASCII.

4. What is Memory location ?

5. Define CD–ROM.

6. Multiply 1101 by 1011.

7. What is duality principle ?

8. How will you represent NANO Gate ?

9. What is Cache Memory ?

10. What are the different types of interrupts ?

11. Define Logical address space.

12. What is cycle stealing ?

13. What is an object program ?

14. What is Batch operating system ?

15. What are the applications of Micro computer ?

Part - B

$(5 \times 5 = 25)$

Answer **all** questions.

16. (a) Explain the term Algorithm. Write an example.

(*Or*)

(b) Write an Algorithm to find the Average of letters in each sentence.

17. (a) What is the purpose of using a parity check list ? Explain.

(*Or*)

(b) Explain the term "data in Machine readable form".

18. (a) What is a speech O/P unit ? What are its Applications ?

(*Or*)

(b) Explain how data is organized on a floppy disk.

19. (a) Write a Machine language program for HYPCOM to find the smallest of three Numbers.

(*Or*)

(b) Show that $(A + B) \cdot (A + C) = A + B \cdot C$

20. (a) What is a page address table ? How it is used ?

(*Or*)

(b) Distinguish between Real time system and Online system.

Part - C (5 × 12 = 60)

Answer **all** questions.

21. (a) How will you represent Hexadecimal, Error detection codes ? Explain in detail.

(*Or*)

(b) Explain the output units with example.

22. (a) Explain the structure of Instructions.

(*Or*)

(b) Evaluate using 2's Complement :

(i) $\begin{array}{r} 5 \\ -3 \\ \hline \\ \hline \end{array}$

(ii) $\begin{array}{r} -5 \\ 3 \\ \hline \\ \hline \end{array}$

(iii) 485 – 128

(iv) 10101 – 10001

23. (a) Explain the Postulates and Duality principles in logic circuits.

(*Or*)

(b) Explain RISC in detail.

24. (a) What is Assembly language ? Explain.

(*Or*)

(b) Explain unix operating system in detail.

25. (a) What are Micro computers ? Explain.

(*Or*)

(b) Write about the classification of computers.

——— *** ———

AFN-1234 PCA132

P.G.D.C.A. (P.G. DIPLOMA) EXAMINATION DECEMBER 2010

First Semester

Computer Applications

OFFICE AUTOMATION

(Non-CBCS)

Time : 3 Hours Maximum : 100 Marks

Part - A (15 × 1 = 15)

Answer **all** questions.

1. How will you move the Windows ?

2. What is File manager ?

3. Define OLE.

4. How will you Edit a document ?

5. How will you check the spelling ?

6. Give any two features of MS-word.

7. “Editing the cells”–Explain.

8. Define addressing modes.

9. “Financial functions”– Discuss.

10. Define Database.

11. Write about any two formatting commands.

12. Define Macros.

13. What is power point ?

14. How will you run a slide ?

15. What are the sound effects in MS-power point ?

Part - B (5 × 5 = 25)

Answer **all** questions.

16. (a) How will you open multiple windows ?

Or

(b) How will you rename files and directories ?

17. (a) How will you use tabs ? Explain.

Or

(b) How will you find and replace a text in MS-word ?

18. (a) Write about Insertion, deleting rows and columns.

Or

(b) Write about creating charts

19. (a) What are the drawings in the tool bar of MS-Excel ?

Or

(b) Define work sheets. Explain in detail about multiple work sheets.

20. (a) Write about power point views.

Or

(b) How will you print a presentation ?

Part - C (5 × 12 = 60)

Answer **all** questions.

21. (a) What is File Manager ? Explain the uses of file Manager.

Or

(b) How will you embed an object ? Illustrate.

22. (a) Write the steps in formatting the text.

Or

(b) Write about Mail Merge.

23. (a) Give the features of MS-Excel with respect to Charts.

Or

(b) List the statistical functions and their use in detail.

24. (a) "Inclusion of Database"—Discuss.

Or

(b) Write steps to create and run a macro.

25. (a) "Powerpoint a Boon"—Explain.

Or

(b) Explain how you will create a presentation and how it is viewed ?

——— *** ———

AFN-1235

PCA133

P.G.D.C.A (P.G. DIPLOMA) EXAMINATION DECEMBER 2010

First Semester

Computer Applications

PROGRAMMING IN C

(Non-CBCS)

Duration : 3 Hours Maximum : 100 Marks

Part - A (15 × 1 = 15)

Answer **all** questions.

1. Getc ( ) is a ———— (input / output) function.

2. Printf (" % f " , x); gives ———— (integer/float/ string) value of x.

3. While loop repeates a set of statements until a condition becomes false.

4. By default, the main ( ) function returns a numeric value to the operating system. (True/False)

5. String is the built-in data type in C. (True/False).

6. ———— function is used to count the number of characters in a string.

7. The ———— escape sequence character gives a beep sound. (\n *or* \t *or* \b *or* None)

8. Structures and unions are same in C.

   (True/False).

9. Queue is a FIFO structure.

   (True/ False).

10. A function which calls itself is called ————.

11. ———— tree traversal algorithms gives post fix notation.

12. A leaf node has ———— number of children.

13. The command for close a file in C is ————.

14. The main ( ) function may have atmost ———— number of arguments.

15. Preprocessor directive is used to define ————.

Part - B (5 × 5 = 25)

Answer **all** questions.

16. (a) Describe scanf ( ) function with its formatting options.

(*Or*)

(b) Explain all the bit-level operations.

17. (a) Compare do–while and while loops.

(*Or*)

(b) Write a program to calculate the total and average of an array of numbers.

18. (a) Explain stack and its operations.

(*Or*)

(b) How to define and use the structures ? Explain.

19. (a) What is recursion ? Explain it with an example.

(*Or*)

(b) Write a program to find the nCr value using recursion.

20. (a) Explain the use of command line arguments.

(*Or*)

(b) Explain the macro definitions.

Part - C (5 × 12 = 60)

Answer **all** questions.

21. (a) Explain all the conditional statements, with examples.

(*Or*)

(b) Explain about constant and variables.

22. (a) Write a program to transpose a given matrix.

(*Or*)

(b) How to define and call the functions ? Explain it with illustrative program.

23. (a) Explain pointer data type and its applications.

(*Or*)

(b) What is binary tree ? How it is created and maintained ?

24. (a) Discuss the tree traversal algorithms with its usages.

(*Or*)

(b) Discuss the working principle of a recursive algorithms.

25. (a) Explain the files and its operations.

(*Or*)

(b) Discuss about conditional compilation.

——— *** ———

AFN-1236 PCA134

P.G.D.C.A. (P.G. DIPOMA) EXAMINATION DECEMBER 2010

First Semester

Computer Applications

VISUAL PROGRAMMING

(Non-CBCS)

Time : 3 Hours Maximum : 100 Marks

Part - A (15 × 1 = 15)

Answer **all** questions.

1. What is DDE ?

2. Define handle.

3. What is DC ?

4. Mention the function that is used to get the current data and time.

5. What is the use of Label box ?

6. What are Constants ?

7. Where will you use Rich Text Box ?

8. Expand OLE.

9. What is Slider ?

10. What is the use of Icons ?

11. Write about Cedit.

12. How will you use CFile ?

13. Define MFC.

14. Expand ODBC.

15. What is the purpose of CMutex ?

Part - B (5 × 5 = 25)

Answer **all** questions.

16. (a) What do you mean by calling Conventions ?

Or

(b) What are windows Components ?

17. (a) Write about Strings.

Or

(b) Explain user defined Functions and Modules.

18. (a) What are Custom Controls ?

Or

(b) Explain the different Graphic functions.

19. (a) Write about MFC class hierarchy.

Or

(b) Explain AFX functions.

20. (a) How the ODBC components are used ? Explain.

Or

(b) Explain the Data Access Methods.

Part - C (5 × 12 = 60)

Answer **all** questions.

21. (a) Write a SDK program for simple window creation.

Or

(b) Explain in detail about new executable and portable files.

22. (a) Write notes on :

(i) Functions.

(ii) Program flow control.

Or

(b) Write a VB program to find the Current date and time.

23. (a) How will you create and manipulate an object in VB ? Explain.

Or

(b) Explain :

(i) DDE properties.

(ii) Data Grid.

24. (a) Explain the MFC Resources.

Or

(b) Explain Document / View Architecture.

25. (a) Describe the steps in creating MDI applications in VC++.

Or

(b) Explain the synchronization classes.

——— *** ———

AFN-1237 PCA231

P.G.D.C.A. (P.G. DIPLOMA) EXAMINATION NOVEMBER/DECEMBER 2010

Second Semester

Computer Applications

OBJECT ORIENTED PROGRAMMING AND C++

(Non-CBCS)

Time : 3 Hours Maximum : 100 Marks

Part - A (15 × 1 = 15)

Answer **all** questions.

1. << is called (insertion/extraction) operator.

2. To set the width of an output, ———— manipulator is used.

3. Default is a reserved word. True/False.

4. & operator is used before the parameter.

5. ——— operator is used to establish a link between class and member function, when member functions are defined outside the class.

6. ——— operator is used to access the members of an object through pointer to the object.

7. Destructors can be passed with arguments. True/False.

8. Operator can't be overloaded. True/False.

9. ——— Function, an outside function can access the private members of an object.

10. Mention the two advantages of inheritance.

11. There are ——— number of polymorphisms.

12. Classes can be nested. True/False.

13. ——— keyboard watches for an exception.

14. ——— function of file stream opens a file.

15. Template functions are also called generic functions. True/False.

Part - B (5 × 5 = 25)

Answer **all** questions.

16. (a) Write the applications of OOP.

Or

(b) Compare OOP with procedural programmig paradigm.

17. (a) Describe the general structure of a class.

Or

(b) Write a function that contains default arguments and call it with omitted arguments.

18. (a) Explain the use of constructor with the help of a program.

Or

(b) Describe unar–y operator overloading through an example program.

19. (a) Write different modes and types of inheritance.

Or

(b) What is virtual base class ? Illustrate.

20. (a) Explain briefly about file I/O.

Or

(b) Write a template function to multiply two numbers of different types.

Part - C (5 × 12 = 60)

Answer **all** questions.

21. (a) Describe software evolution and object oriented programming concepts.

Or

(b) Explain the use of control structures in C++, giving examples.

22. (a) Write a sample C++ program to illustrate call by reference and call by value mechanisms of calling functions.

Or

(b) Explain in detail about function overloading concept and rules governing it.

23. (a) Explain the use of the following :

(i) Copy Constructor. (6)

(ii) Dynamic Constructor. (6)

Or

(b) Write a C++ program for student information system with parameterized Constructor in student class. Write necessary functions.

24. (a) Write a C++ program to illustrate hybrid inheritance.

Or

(b) Explain the order of execution of Constructors and Destructors in Multilevel inheritance.

25. (a) Write a C++ Program to illustrate the usage of various I/O manipulators.

Or

(b) Explain Exception handling facilities in C++ through examples.

——— *** ———